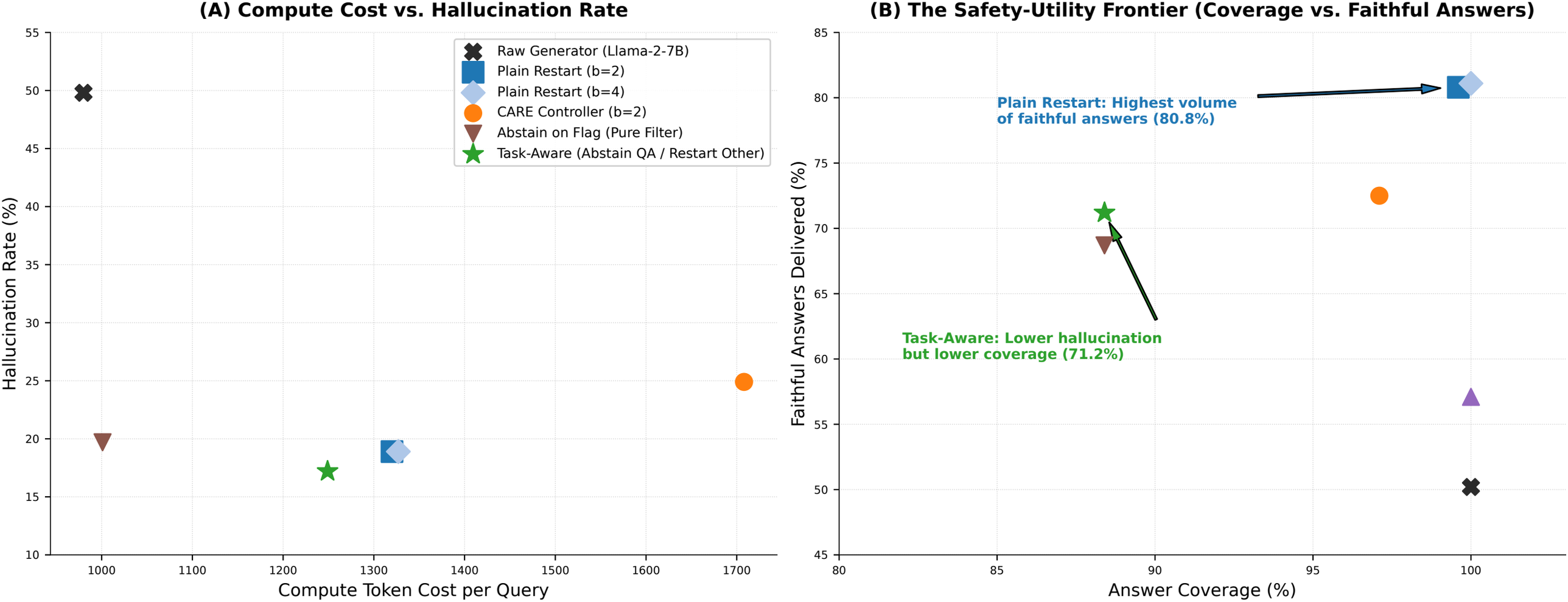
(A) Compute Cost vs. Hallucination Rate
Hallucination Rate (%)
Compute Token Cost per Query
Raw Generator (Llama-2-7B)
Plain Restart (b=2)
Plain Restart (b=4)
CARE Controller (b=2)
Abstain on Flag (Pure Filter)
Task-Aware (Abstain QA / Restart Other)
(B) The Safety-Utility Frontier (Coverage vs. Faithful Answers)
Faithful Answers Delivered (%)
Answer Coverage (%)
Plain Restart: Highest volume
of faithful answers (80.8%)
Task-Aware: Lower hallucination
but lower coverage (71.2%)